

Two decisions, two clocks

DECISION A • WEEKS • FIRM

Bravo's runtime is out of support with a live code-execution path. Fund the engine and framework upgrade now — 18–33 engineer-days, no licence.

DECISION B • MONTHS • OPEN

Where the next product family is built is genuinely open. Three measurements gate it and none has been taken.

Separating the two decisions is the most useful thing in this pack

DECISION A – URGENT		DECISION B – STRATEGIC
The question	How do we make Bravo safe to run while B is taken?	What is Bravo’s platform for the next 5–10 years , and where does the next product family go?
Clock	Weeks	Months
Answer	Firm. Swap the engine and upgrade Spring Boot in one change – 18–33 engineer-days, Rp45–125M, no licence	Open. Three measurements gate it, none taken
Contingent on the other?	No	No

Decision A is unconditional under every strategic path.

Every option leaves Bravo running the retail book for at least 12 months — a Temporal port takes 12–18, a LORA migration 15–24. And the Spring Boot half of the work is owed under those paths anyway. Nothing in the rest of this deck should delay it.

WHY A CANNOT WAIT FOR B

Bravo is not approaching two end-of-support cliffs. **It has gone over both**, and it sits on the last publicly available build of each stack — with 61 injected remote-code-execution process definitions in the production engine and no patch channel to fix them.

WHY B IS OPEN RATHER THAN RECOMMENDED

The previous version of this brief recommended prioritising LORA. **Three of the five pillars that rested on have since failed** — set out on the *Evidence base* slide. What survives is genuinely balanced, and the evidence that would decide it does not exist yet.

WHAT IS TRUE EITHER WAY

Bravo carries the majority of the book and the majority of the money — ≈Rp 50.2bn against ≈Rp 2.4bn. It must be resourced and treated as strategically important regardless of where new products go, because it will be originating most of BFI’s business value throughout whatever migration is or is not chosen.

Bravo runs on two dead foundations, and one of them is being probed

COMPONENT	BRAVO RUNS		SUPPORT ENDED
Camunda 7 Community Edition	7.23.0	14 Oct 2025 – line closed, final artifact 7.24.0, repo archived, no security patches ever again	
Spring Boot 3.5.x (OSS)	3.5.16		30 Jun 2026
Java 17 (Oracle premier)	17		30 Sep 2026
Camunda 7.23 (Enterprise maintenance)	– no licence held –		13 Oct 2026
Verified in the checkout: pom.xml:26 pins 7.23.0; no Camunda BOM, no enterprise artifact, no licence file or property, no Camunda EE repository – only BFI’s own Artifact Registry.			

AND THE ENGINE HAS ALREADY BEEN USED TO RUN CODE IN PRODUCTION

Both the `-webapp` (Cockpit/Tasklist/Admin) and `-rest` starters are **embedded in the service**. The production engine contains **61 injected remote-code-execution process definitions**, deployed across 7 sessions between 2026-05-23 and 2026-06-30, **one confirmed executed** (it captured the pod hostname), **all still active**.

Vector corrected 2026-09-10, after the Bravo team pushed back: the deploy path `/engine-rest/**` *does* require a credential, so this is a **compromised or shared secret, not an open door** — and the shared `INTERNAL_SERVICE_KEY` grants full engine rights. Worse in one way: someone held a valid BFI credential.

An engine with no patch channel and a publicly reachable deployment endpoint is the worst combination on the list.

And it is live today. This is why Decision A is separated out and why it does not wait for anything else in this deck.

A supported destination exists, and it is cheap

Two Apache-2.0 community forks of Camunda 7 run on Spring Boot 4, keep the `ACT_` schema unchanged, accept the legacy `camunda:` BPMN namespace and ship automated migration recipes. Swapping the engine **and** upgrading Spring Boot in one change is **18–33 engineer-days, Rp45–125M, no licence** — against 31–57 engineer-months for a Temporal port or a LORA migration. **The 53 BPMN files and Java 17 are unchanged.**

OPERATON 2.1.4		CIB SEVEN 2.2.0
Commits, last 52 weeks	2,310	258
Distinct authors, 12 months	52	31
Top-contributor concentration	~80%	25%
Ex-Camunda core engineers active	none	two
Long-term support	none — 6-month lines	paid, terms undisclosed
Legal entity	none yet; non-profit planned	CIB software GmbH
Requires a 7.24 hop first	yes	no
Cockpit webjar path	moves	unchanged

THE ONE OPEN SUB-DECISION

Is a contractual support agreement **required** for the workflow engine — and can one actually be purchased for a BFI entity?

Yes → CIB seven. No → Operaton.

START THE ENQUIRY BEFORE DECIDING

CIB seven’s support terms are **not published** — length, service levels and pricing are all sales-contact-only. If the answer may be yes, open that enquiry now rather than after.

ONE RISK TO RETIRE FIRST — HOURS OF WORK

Two Flyway migrations hand-replayed Camunda’s 7.20→7.22 upgrade DDL with `IF NOT EXISTS` guards, so the columns arrived but `ACT_GE_SCHEMA_LOG` may still carry a pre-7.22 version label. **If it does, the engine fails to start at cutover.** Four SQL queries per environment settle it. Run them before setting a date.

What Decision B is — and what it is not

YOU ARE DECIDING THIS

- **Observability.** Bravo emits **zero spans** for its entire orchestration layer and has **zero process-level monitors** among 38.
- **Testability.** Bravo's only scheduled journey run **cannot report a failure**, and the suite behind it has been frozen since November 2023.
- **Whether adding surface to an unwatched platform is acceptable** — each new product family adds more of it.
- **Which platform BFI can actually hire for**, if capacity is added.
Java/Spring/Camunda is a commodity skill set; Go plus a bespoke GSM planner is not. **This is the one staffing-adjacent fact a CTO cannot reverse by reallocating people** — and it favours Bravo.

YOU ARE NOT DECIDING THIS

- **BPMN versus GSM.** Assessed in full — and the **next two slides show the work** rather than asking you to take it on trust: five rationale claims tested against the code, what each paradigm did better, and the Bravo team's review. The paradigm shaped **~6% of Bravo's codebase and ~10% of LORA's**, and favoured LORA there; the other ~90% turns on engineering discipline. Real, settled, and not load-bearing for *this* decision.
- **Whether Bravo works.** It does. 76,446 applications in August, bounded failure, no zombie-loan class.
- **Whether to retire Bravo faster.** Explicitly not recommended.
- **Which platform is cheaper.** Bravo is, by 4–5× per application. Settled.
- **How many engineers each platform has.** That is *your* lever, not a property of either platform — it is the output of the previous decision to consolidate on LORA, and it has been removed from the evidence in this deck.
- **Whether Bravo's engine gets fixed.** That is Decision A, and it is already answered.

A paradigm is fairly judged by what teams build with it under delivery pressure — not by what it permits in principle.

The Bravo team's own review established that Bravo's flowchart problems are consequences of **how Bravo was modelled** rather than laws of BPMN, and that the standard remedies exist. That review is accepted. It does not change the finding, because those remedies are respectively 5.5% **deployed, not started, and not enforced**.

BPMN versus GSM: what the code actually showed

The previous slide says this is *not* what you are deciding. That is a conclusion, and you should see the work behind it rather than take it on trust. LORA’s design rationale made **five arguments against the workflow approach**. Bravo is the concrete workflow system it was arguing against, so each one was testable against the code.

THE RATIONALE CLAIMED	WHAT THE CODE SHOWED	VERDICT
Flowcharts need an arrow for every edge case; complexity explodes	<code>ndf4w.bpmn</code> alone: 111 service tasks, 21 user tasks, 137 exclusive gateways, 48 sub-processes, 70 error definitions in one 9,494-line file, with product routing inside gateway condition strings	● LORA
Loan events are parallel and out-of-order; flowcharts handle it badly	Bravo is almost entirely sequential: 14 parallel and 4 inclusive gateways across 53 files , and zero message events, signal events, event-based gateways or event sub-processes. Camunda supports all of them – their absence is a modelling choice, not an engine limit	● LORA
Workflow data “lives in scattered process variables outside the diagram”	Not how Bravo did it. Camunda holds only routing flags; business data is a relational aggregate of 238 JPA entities, which buys SQL fleet queries. What <i>is</i> scattered is lifecycle state – 30 <code>ApplicationStatus</code> values plus ~105 other status enums and 197 <code>setStatus</code> sites	● REFUTED
Temporal gives retries and durable waits “natively”; engines need plumbing	Bravo built plumbing – and got a better retry policy for it : 186 retry-cycle declarations, a last-attempt degrade framework, a DB-backed retry counter, a transactional outbox and an out-of-order inbox	● BRAVO
Saga compensation for a failed disbursement is the execution layer’s job	Bravo: 0 compensation events, 0 BPMN transactions. LORA: document-field rollback only. Failed go-live is handled by a 5-minute sweeper that re-requests and asks CONFINS to republish	● NEITHER BUILT IT
Full assessment with every file and line cited: <code>compare-architecture.md</code> §2 and §3.1–3.14.		

THE CLAIM THE RATIONALE SHOULD HAVE MADE AND DID NOT

Human tasks re-create flowchart complexity regardless of paradigm. Bravo spends 44% of its code on human work (a 10,419-line surveyor assignment service); LORA’s own assessment found ~12k lines of hand-written state machines inside the declarative shell (`survey.go` : 5,473 lines, 125 transitions). Same phenomenon, both paradigms – and it is where both systems spend most of their code.

Why it is not the load-bearing argument

THE VERDICT ON THE RATIONALE

The paradigm decided the shape of ~6% of Bravo’s codebase and ~10% of LORA’s — the automated pipeline — and decided it in LORA’s favour. The other ~90% looks structurally similar in both, and the differences there come from engineering discipline, not GSM versus BPMN.

BRAVO (PURE WORKFLOW) DID BETTER AT	LORA (GSM + TEMPORAL) DID BETTER AT
– Bounded, classified failure — no zombie-loan class by construction. A claim about the <i>shape</i> of failure, not its frequency – A designed dead-letter path — the “visible failure” LORA’s own reliability doc asks for – Fleet queries — “all loans stuck at survey” is a <code>WHERE</code> clause – A forced retry policy — an engine that makes you write one ends up with a better one than an engine that lets you skip it – Orchestration-tier cost — ≈Rp58M/month against ≈Rp431M	– Adding an automated step — 172 activities, 3 hard precursors, no orchestration edit. The clearest validated win – One artefact — the whole loan in one schema-validated document with a version chain – Parallelism for free — ReadSet/WriteSet locking against Bravo’s 5 parallel gateways in 53 files – Product isolation at the data layer — separate documents and queues; Bravo has one job executor for everything – Observability of the pipeline — a span per activity attempt; Bravo has none

THE BRAVO TEAM REVIEWED THIS, AND THEIR REVIEW WAS ACCEPTED

Four responses were checked against the code (§8). **None moved the verdict, two sharpened it, and the fourth reversed a cost claim in Bravo’s favour** — like-for-like, Bravo’s orchestration tier is the cheaper one. Their central point is **correct and accepted**: the flowchart explosion and the sequential flows are consequences of *how Bravo was modelled*, not laws of BPMN, and the standard remedies exist. It does not change the finding, because those remedies are respectively 5.5% **deployed, not started, and not enforced** — and a paradigm is fairly judged by what teams build with it under delivery pressure, which is the only condition under which either will ever run.

The scoreboard 1 / 2

BRAVO LOS		LORA	EDGE
PEOPLE			
Active engineers	25 across 4 LOS repos (was quoted as 7 – one project only)	22	● EXCLUDED – YOUR LEVER
Repositories touched per integration change	1, or 2 where a console is involved	5, in a mandated order	● BRAVO
Delivery tickets, Jan-Aug	6,518 (BLCS 2,813 + LN 3,705)	5,144	● EXCLUDED – IT INVERTED ON A SCOPE FIX
Median ticket cycle time, Aug	2 days	2 days	● TIE
Hiring pool (not CT0-reversible)	Java / Spring / Camunda – commodity	Go + a bespoke planner + Temporal	● BRAVO
TESTING			
Test files	2,286 across 4 repos (1,457 ms-bpm + 829 consoles)	608 across 13 repos; 4 repos at zero	● BRAVO
Unit tests block a merge	yes – no true in 28 console workflows	yes, where a workflow exists	● TIE
Coverage enforced, not just measured	no – thresholds at 0% and 1%; bumper installed, never invoked	no	● TIE – SAME DEFECT
Automated proof a loan can still be originated	595 tests, frozen 2023; the one scheduled run cannot fail	nightly exists and asserts – mostly red	● LORA
Can a test make an upstream fail on demand?	no – 50 error handlers, 70 error definitions, 190 escalations unexercised	yes – a 2-line delta per test	● LORA
Cost of closing the test gap	L0+L1 run in seconds, ~7 days of work	CI runner has nowhere to run; SIT access unresolved	● BRAVO
OBSERVABILITY			
Spans on the orchestration layer	zero	one per activity attempt, carrying the loan id	● LORA
Can you join a failure to a customer?	no	yes – the workflow id is the application	● LORA
“All loans stuck in survey?”	yes – one SQL query	cannot be asked at all	● BRAVO
Which upstream is failing?	yes – 25 hosts, structurally	no – 300 proxies on one route	● BRAVO
Continued on the next slide: cost, delivery, reliability, volume and platform lifecycle.			

The scoreboard 2/2

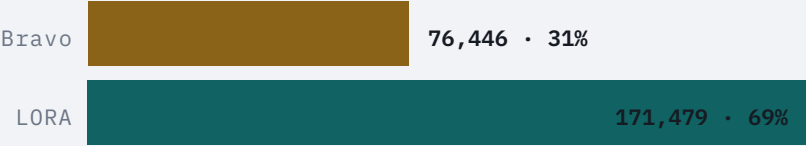
	BRAVO LOS	LORA	EDGE
COST			
Orchestration tier, monthly	≈Rp 58M	≈Rp 431M	● BRAVO 7×
Cost per application	≈Rp 510-760	≈Rp 2,500-3,200	● BRAVO 4-5×
Marginal cost of one more application	Pods + database – small but real	≈zero	● LORA
DELIVERY			
In-flight deploy safety	a sub-process redeploy reaches in-flight loans – 73% of volume	version-pinned queues	● LORA
Front-end errors, 7 days	2,413,198 across 4 consoles	416,201 on one back office	● LORA
Shared UI library kept in sync	no – one design system, three versions across the three LOS consoles; React 17 vs 18, Vite 4 vs 7, ESLint vs Biome	one back office, one toolchain	● LORA
Master data → dropdowns	live – a role rename was 1 YAML line + 1 SQL update	startup cache, copied onto the loan	● BRAVO on speed
RELIABILITY			
Manual intervention rate	0.443% – 1 in 225	0.126% – 1 in 790	● LORA 3.5×
Failure shape	bounded, classified, parked – no zombie class	uncapped retry; ~half never terminate	● BRAVO
VOLUME & BUSINESS VALUE			
Applications, Aug 2026	76,446 (31%)	171,479 (69%)	● LORA ON COUNT
Business value	≈Rp 50.2bn	≈Rp 2.4bn	● BRAVO ≈21×
Value per application	≈Rp 657,000	≈Rp 14,000	● BRAVO ≈47×
Product families live in production	4 – NDF2W, NDF4W, RO, DF4W	1 – NDF (2W 90.2%, 4W 9.8%)	● BRAVO
Families built, not yet live	DF2W – in UAT, LOS pen test running, go-live epics open	6 of 7 – awaiting a CTO decision; three with no deploy-prod.yaml	● BRAVO
PLATFORM LIFECYCLE			
Workflow engine support	Camunda 7.23.0 CE – dead 14 Oct 2025, no licence held	Temporal Cloud, contracted	● LORA
Framework support	Spring Boot 3.5.16 – dead 30 Jun 2026	Go, current	● LORA
Cost to reach a supported runtime	18-33 engineer-days, no licence	n/a – already supported	● DECISION A
Full matrix with per-row sources in compare.md §1. Rows where both platforms are equally weak are collected on the What converges slide.			

Applications were the wrong unit, by a factor of about 47

Every rate in this analysis divides by an **application count** from a billing sheet. On that measure LORA carries 69% and Bravo 31%, and the previous version of this brief used it as a reason to prefer LORA.

Measured by business value the ratio inverts.

APPLICATIONS, AUG 2026



BUSINESS VALUE



VALUE PER APPLICATION

Bravo ≈ Rp 657,000 LORA ≈ Rp 14,000

LORA IS LIVE ON ONE PRODUCT FAMILY, NOT SEVEN

The prod cluster runs **only** the NDF workers. The live set is **NDF2W (90.2% of loans) and NDF4W (9.8%)**, both through the partnership door. **Six of seven families are un-deployed** — three have **no production deploy workflow at all**, and three more have a copied one naming a schema version that does not exist. There is no switch to flip: enabling a family is a deploy-and-launch project.

So the previous claim — “*already carries 69% of applications across seven product families*” — was wrong. **That 69% all runs on the one family.**

THIS PACK APPLIED ITS OWN CRITICISM
ASYMMETRICALLY – THEN OVER-CORRECTED

Our analysis faulted Bravo for DF2W being “*fully configured, zero applications*” while crediting LORA for **six families in that state**. That double standard was real. But the two are **not the same state**: DF2W is in UAT with a live pen test and a dated go-live path; LORA’s six await a decision and three cannot deploy at all. *On this row the edge is Bravo’s*. What survives: unlaunched families tell you nothing about ease of building, on either platform.

WHAT IS MISSING ENTIRELY

No agreement, disbursement, NTF or revenue data exists anywhere in either platform’s records. The billing sheet has 14 cost columns and 2 application-count columns. The Rp figures above are a **CTO-office estimate, period and basis to be confirmed** — and since the ~47× ratio is load-bearing, confirming it is gating measurement #1.

Strip out what ties, what is your own lever, and what cancels. Three measurements remain.

4–5_x CHEAPER PER APPLICATION ON BRAVO – AND IT CARRIES ~21_x THE VALUE	3.5_x MORE OFTEN BRAVO NEEDS A PERSON – BUT ONE TICKET CATEGORY FROM LEVEL	0 SPANS, PROCESS MONITORS AND WORKING JOURNEY RUNS ON BRAVO’S ORCHESTRATION	3 MEASUREMENTS THAT WOULD SETTLE THIS – NONE TAKEN
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1 · COST AND VALUE – POINTS AT BRAVO

≈Rp58M against ≈Rp431M, 4–5_x cheaper per application, and ≈21_x the business value. **Bounded two ways:** LORA’s gap is fixed contracts and over-provisioning, not the paradigm; and Bravo’s unit advantage erodes on its own as volume drains — ≈Rp1,450 per application at 40k/month.

2 · INTERVENTION RATE – POINTS AT LORA, ONE CATEGORY FROM A WASH

0.443% against 0.126%. But the gap **turns entirely on** **Surveyor Platform - Release reject** — 28.4% of Bravo’s load, growing 4.8_x, **still unmapped to any code path**. Remove it and Bravo is at 0.152% against 0.126%. **Days of work. Gating measurement #2.**

3 · OBSERVABILITY AND TESTABILITY – POINTS AT LORA, AND IS CHEAP TO FIX

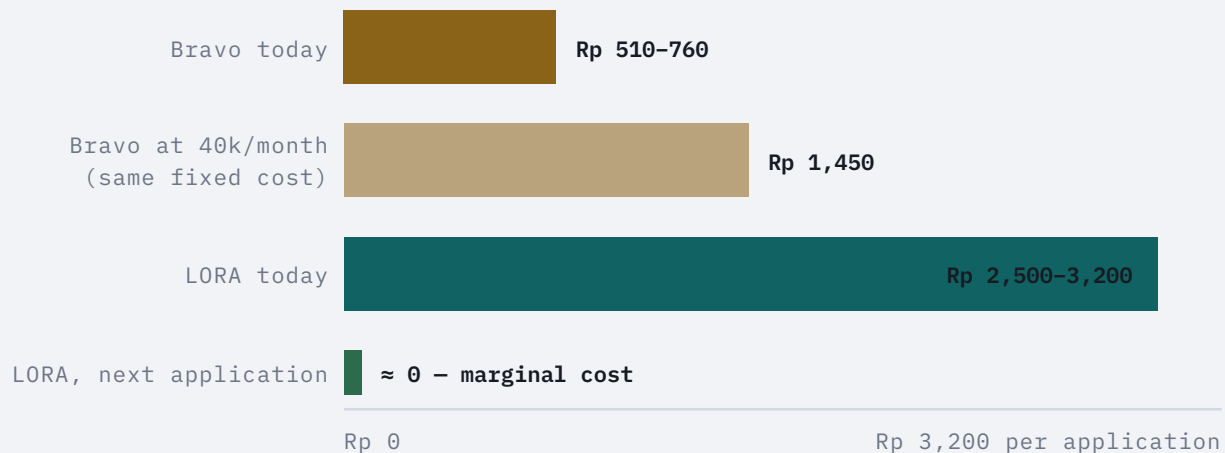
Zero orchestration spans, no loan id in logs, zero process monitors among 38, a journey suite frozen with its scheduled run unable to fail. **But the two layers that would catch the most run in milliseconds** and are ~7 days of work, and 349 usable journey descriptions already exist. **A decision nobody has made, not a mountain nobody can climb** — so it is a reason to fund remediation, and a weak reason to move products.

TWO DRIVERS WERE WITHDRAWN SINCE THE LAST VERSION, AND BOTH HAD POINTED AT LORA

Team size — 7 against 22 — is the output of the previous decision to consolidate on LORA, reversible by the same office. Circular as evidence. **And “Bravo’s modern generation is only 5.5% of its volume”** cannot carry weight while LORA’s entire production estate is one family and six of its seven have never launched. *Symmetrically applied, it cancels* — and DF2W, the family that would raise Bravo’s share, is in UAT now.

Bravo's strongest argument is cost — and it erodes without anyone acting

This is the one place the evidence favours **Bravo** decisively, and it should not be minimised: **~Rp58M against ~Rp431M** for the orchestration tier, and **4–5× cheaper per application**. One deployment, one database, against fixed contracts and over-provisioned capacity.



WHY THE GAP IS NOT THE PARADIGM

LORA's cost is ~86% fixed and its levers are **time-locked, not architectural**: a 3-year CPU commitment, a Temporal commitment renegotiable at ~**March 2027**, 448 GB of requests running at 15.5% utilisation, and five idle worker versions. **Provisioning, not physics.**

WHY IT ERODES ON ITS OWN

Bravo's platform cost is fixed and its volume is falling *by design* — **-35% in one month**. At 40k applications the same $\approx$ Rp58M is **$\approx$ Rp1,450 each** and the gap halves with nobody doing anything. Meanwhile LORA absorbed **+23% volume for +3% spend**: the next product is close to free to host.

AND THE BIGGEST COST LEVER IS UNRELATED TO THE DECISION

Cloud Logging alone is Rp140.5M/month – 2.4× the entire tier the migration would retire – and it is a configuration fix behind a global Feign setting. A team that fixed logging config would save more in September than retiring Bravo LOS will save when the migration completes. Do it either way.

Observability and testability — the two that are real, and both are cheap to fix

OBSERVABILITY — THE PROCESS IS INVISIBLE

Zero Camunda spans. Seven operation names cover every span the service emits and none of them is a process, a service task or a job execution — because all 483 service tasks run on job-executor threads that no HTTP request started. **The 90% of Bravo's work that is the loan moving through the flow produces no trace at all.**

No log line carries the application id, so a failure cannot be joined to a customer without leaving Datadog. **38 monitors, zero on the process**; 8 permanently red, 6 that cannot fire. A health model of “not 5xx” missed **748,686 404s a week reaching 69% of surveyor sessions.**

TESTABILITY — THE PROOF WAS SWITCHED OFF

Bravo wrote **595 Cypress journey tests** across 46 authors and aimed **349 of them at the Surveyor Platform** — the console generating 28.4% of its tickets. Then it froze the repository in **November 2023** and left the one scheduled workflow with **every step ending `|| true`**, so it can never report a failure.

An absent test cannot mislead anyone. A permanently green one can. And no Bravo test can make an upstream fail, so **50 error handlers, 70 error definitions and 190 escalation paths have never been executed** — including an anti-fraud bypass that is a credit-policy decision reached by an exception handler.

One point in Bravo's favour: the corpus's **130 API feature files assert on 87% of their `Then` steps** — competent Gherkin, and the best-written in the estate. The defect is confined to the 465 UI files, where 57% of steps are pure clicking and typing. **The skill is present; it was not applied where it mattered and then it was switched off.**

Each new product family adds surface, and on this platform nothing watches it.

Capacity used to be the third panel here. It has been removed: **headcount is the output of the previous decision to consolidate on LORA**, so citing it to the office that holds that lever is circular. What is left is a statement about **engineering discipline** — and it is fixable in days-to-weeks, which is why it is a reason to fund remediation rather than to move products.

Seven things are identical on both platforms — so none of them chooses for you

This slide exists to keep the decision honest. **Anything that is the same on both cannot be a reason to prefer either**, and these seven are the strongest evidence that the recommendation is about engineering practice rather than architecture.

	BRAVO	LORA
Human-task complexity returned in both	44% of the codebase is human work	~12k lines of hand-written state machines inside a declarative shell
The orchestration layer is untested in both	4 of 2,286 tests execute a process	planner scheduling untested; nightly red
Both built a journey suite and left it unable to run	595 features, 46 authors, frozen 2023, scheduled run cannot fail	<code>lora-super-test</code> : 150 tests, a CI workflow with nowhere to run
The lifecycle state machine is declared, not enforced	a state-machine class used once	enum membership validated only
Neither has saga compensation	sweepers, re-sync, humans	sweepers, re-sync, humans
Both mistake one absent failure mode for health	“not 5xx = healthy”	“still Running = healthy”
Both filtered their most informative error out of their one business monitor	excluded <code>ENGINE-16004</code>	excluded the go-live agreement message
Surveyor assignment is the top ops complaint on both	1,182 tickets – 21.8%	1,021 tickets – 25.0%

Two teams, two paradigms, the same outcome — repeatedly.

A BPMN flowchart and a GSM planner each modelled surveyor assignment around a hand-written service layer and inherited its failure modes, at nearly the same rate. Neither team knew the other had abandoned a journey suite.

THE CHEAPEST ITEM ON EITHER BACKLOG

Both teams had rich instrumentation they were **not reading**. LORA's two RUM applications had been on all along and were never consulted; all four Bravo consoles are instrumented and produce **2.4M errors a week that nobody had opened**. This is not an architecture finding. It is the same operational gap on both sides, and it costs a weekly meeting to close.

What you are accepting, whichever way you decide

IF THE NEXT PRODUCT GOES TO BRAVO

- **You are committing to a community fork of a dead engine.** Decision A lands Bravo on Operaton or CIB seven — maintained, but on a 6-month support line or an unpublished paid contract. Today’s exposure is *drift*; that would be a *chosen* dependency, and it attracts different scrutiny.
- **The new product will be as unobservable as the existing ones.** Zero orchestration spans, no application id in logs, no process-level alert — so its failures surface as support tickets.
- **It inherits five product-discrimination mechanisms.** Adding a product touches all five, and no test or CI artefact maps a change to the products it affects.
- **It inherits the in-flight deploy hazard.** A shared sub-process redeploy reaches loans already running — today that is 73% of volume with no migration plan.
- **You get a genuinely cheaper tier** and the lowest per-change cost in the estate. That is the trade.

IF THE NEXT PRODUCT GOES TO LORA

- **You pay a five-repository tax per integration change** — schema, handler, activity, authorisation, front end, in a mandated order. Real, measured, and fixable with a generator that is designed but not built.
- **You inherit uncapped retry** and a class of loans — roughly half — that never reach a terminal state while every signal reads healthy. LORA’s reliability edge is *partly* an artefact of failures nobody can see.
- **You cannot ask fleet questions about business state.** “All loans stuck in survey” has no answer in LORA at all.
- **You cannot tell which upstream is degrading** until a merged PR is actually released.
- **You are betting on a platform proven on one product family.** Six of seven are un-deployed, three without a production deploy workflow — so “LORA hosts seven families” is a statement about repositories, not production.
- **You cannot yet show that LORA converts.** No agreement, disbursement or revenue data exists for it. On the CTO office’s figures it earns ≈Rp 14,000 per application against Bravo’s ≈Rp 657,000.
- **You get near-zero marginal cost**, product isolation by construction, version-pinned deploys, and a journey test that asserts. That is the trade.

Both columns are now roughly the same size, which is the honest position: LORA’s risks are unproven scale, Bravo’s are unfixed discipline, and both are addressable with engineering that has already been scoped. That is why Decision B is presented as open rather than answered.

Three measurements gate this. None has been taken.

#	MEASUREMENT	WHAT EACH OUTCOME WOULD MEAN	COST
1	The agreement / NTF / disbursement split by platform, from a system – with a stated period and basis	Confirms $\approx 21\times$ → Decision B tilts to Bravo for anything revenue-bearing and LORA’s volume argument disappears. Shows comparable conversion → tilts back to LORA . Same measurement, opposite outcomes – which is exactly why it must be taken rather than assumed	days
2	Surveyor Platform - Release reject mapped to a code path Narrowed 2026-09-10: six named candidate endpoints – four release-assignment handlers in <code>OperationAssignmentController</code> , plus <code>voidAssignment / reprocess</code> and the surveyor console’s <code>cancel-reject-notes</code> routes. Squad LN.	If it is a fixable defect, Bravo’s intervention rate falls to $\approx 0.152\%$ against LORA’s 0.126% – level – and LORA’s clearest remaining reliability edge disappears	hours
3	The cost of launching one un-deployed LORA family – three of the six have no production deploy workflow	Prices the actual proposal. “Put the next family on LORA” assumes launching a family there is cheap, and nobody has done it since NDF	weeks

AND THESE WOULD MOVE IT WITHOUT BEING COMMISSIONED

IF THIS TURNS OUT TO BE TRUE	EFFECT
Decision A is funded and lands	Removes the runtime-support and RCE exposure from Decision B entirely, so B can be taken on architecture and economics alone. 18-33 engineer-days. The one item here that should not wait for evidence.
The billing sheet’s application counts are defined and Bravo’s is confirmed inflated	Bravo’s true intervention rate and per-application cost both rise; its value-per-application rises further still
<code>bravo-e2e-test</code> ’s weekly cron turns out to still be firing	Bravo has had a green-by-construction journey signal for three years on 10 of 595 files. A misleading signal is worse than none. Five minutes in the Actions tab
LORA’s never-terminating loans and uncapped retry go unfixed for two more quarters	LORA’s reliability edge is partly failures nobody can see
Bravo is re-staffed	No longer a falsifier. Headcount was withdrawn as evidence – it is an input you set, not a finding that moves

One decision to take now, one to defer, four things to authorise

DECISION A · TAKE IT NOW

Fund **Option 1 Path B** — fork the engine and upgrade Spring Boot in one change. **18–33 engineer-days, no licence. And fix the RCE this week.**

DECISION B · DEFER, DELIBERATELY

Do not place the next product family until measurements 1 and 2 are in. They cost days, and either could decide it.

1	Fix the Camunda RCE this week. Rotate <code>INTERNAL_SERVICE_KEY</code> — a single static shared secret that grants full engine rights, so it is RCE-equivalent — and the plaintext secrets committed in <code>bravo-e2e-test</code> . Identify the caller from <code>POST /engine-rest/deployment/create</code> access logs (2026-05-23 → 06-30). Register a <code>ProcessEngineAuthenticationFilter</code> so the engine's own authorization applies. Purge the 61 definitions; remove <code>permitAll()</code> on <code>/camunda/**</code> as hardening; check for lateral movement.	THIS WEEK REQUIRED UNDER EVERY OPTION
2	Fund the engine + Spring Boot upgrade. Answer the fork question first — is a contractual support agreement required and purchasable? Then run the <code>ACT_GE_SCHEMA_LOG</code> diagnostics before setting a cutover date.	18–33 ENG-DAYS RP 45–125M NO LICENCE
3	Commission the agreement / NTF split by platform , with a stated period and basis. The Rp 50.2bn / Rp 2.4bn figures are a CTO-office estimate and the ~47× ratio is load-bearing. Pair it with defining <code>Bravo total app</code> and <code>Lora total app</code> .	DAYS GATES DECISION B
4	Authorise the Cloud Logging fix — Rp 50–90M/month, configuration only. Larger than the saving from retiring the whole Bravo LOS tier, available now, and it stops PEFINDO, SLIK, Dukcapil and CONFINS request bodies landing in Cloud Logging. Unconditional.	WEEKS RP 50–90M / MO

AND ONE STANDING INSTRUCTION THAT DOES NOT WAIT FOR ANY OF IT

Bravo carries the majority of the book and the majority of the money. Whatever is decided about new products, it must be resourced and treated as a strategically important platform — not as a system being wound down. Its runtime is out of support today, its orchestration is unobservable, and its journey tests do not run — and it will be originating most of BFI's business value throughout whatever migration is or is not chosen.

The finding that makes Decision A urgent rather than tidy

CAMUNDA RCE EXECUTED ON PRODUCTION BPM VIA AN AUTHENTICATED DEPLOY PATH

The production `ms-bpm` Camunda engine contains **61 process definitions that are not part of the application**, deployed across **7 distinct sessions between 2026-05-23 and 2026-06-30**. They are textbook expression-injection remote-code-execution probes — reflection chains that build `java.lang.Runtime` through string `concat()` to dodge keyword filters, then run OS commands and capture stdout into a process variable.

All **61 are still active and startable today**, and there is evidence in the engine's own history that **at least one executed**. Camunda Cockpit is `permitAll()` at the Spring Security layer — but **corrected 2026-09-10, that is not the vector**. The deploy path is `/engine-rest/**`, which requires a Keycloak token or the shared `api-secret`; the shared secret grants full engine rights, and no `ProcessEngineAuthenticationFilter` is registered, so Camunda's own authorization never applies. **Someone held a valid credential.**

This was found incidentally while querying the production database for the workflow analysis. If it was an authorised penetration test, treat this as confirmation the artefacts are still in production. **If it was not, this is an incident.**

61

INJECTED RCE PROCESS DEFINITIONS, ALL STILL ACTIVE IN PRODUCTION

This does not bear on where the next product goes.

It is on this deck for one reason: it needs an owner and a route to whoever holds Bravo security, and a platform-strategy decision is exactly the kind of thing it could get parked behind. **It should be actioned this week, independently of decision 1.**

WHY IT IS HERE RATHER THAN IN THE SCOREBOARD

Including it as a Bravo mark against LORA would be misleading — it is an access-control and deployment-hygiene failure, not a property of BPMN, and nothing in this analysis establishes that LORA's equivalent surfaces are better protected. **They have not been checked. That is itself worth an hour.**

Where every number on these slides comes from

MEASURED FOR THIS ANALYSIS

Jira — projects `DF` , `D2W` , `BL` , `BLCS` ; four 50-issue samples across two months.

Datadog us5 — APM spans, 1.1M log lines/week, the full monitor inventory, Synthetics, CI Visibility, and RUM on four production consoles.

GCP billing via the FinOps API, complete month August 2026, per-project and per-service.

Both codebases — `bravo-bpm-service` at v2.93.43, `bravo-e2e-test` , and 13 LORA repositories.

OTRS ticket export for the symmetric reliability measurement.

KNOWN WEAKNESSES, STATED

The denominators. Application counts come from a billing sheet whose platform totals nothing else corroborates. Every per-application figure is a verified numerator over a disputed denominator — ask #2 fixes this.

The drain confound. Bravo keeps the residual hard cases as it is migrated. Real, and not controlled for.

One category carries driver 2. Now with six named candidate endpoints and an owning squad (`LN`), so this is hours of instrumentation rather than an open search. `Release reject` is 28.4% of Bravo's tickets and unmapped — ask #1.

Two database MCP servers failed to connect during the final session, so SQL-shaped claims are marked *mechanism verified*, *volume not re-measured*.

Four corrections already made, three of which removed arguments this brief used to rely on:

- the first version said Bravo had no journey suite; it has 595 tests, frozen. That correction was *worse* for Bravo.
- **team size was used as evidence against Bravo.** It is the output of the previous decision to consolidate on LORA, so it was circular. Withdrawn.
- **“LORA carries 69% of applications across seven product families” was false.** One family is live. Withdrawn.
- **Bravo's scope was too narrow.** Three console repositories (763,861 LOC, 829 gating tests) and the `LN` delivery project were never counted. Test files went 1,457 → 2,286; engineers 7 → 25; the ticket ratio *inverted*.

The pattern behind all four: Bravo was bounded at one repository and LORA at its whole workspace. Every correction that followed moved toward Bravo. The findings that survived all four — the engine end-of-support exposure, orchestration blindness, the cost ratio, the intervention-rate gap — are the ones to trust.

READ FURTHER

`compare.md` — the 40-row matrix with per-row sources, and this recommendation in full.

`compare-architecture.md` — BPMN vs GSM+Temporal at the paradigm level, plus the Bravo team's review and the reply to it.

Five findings documents — people, testing, observability, cost, delivery. Each carries its own 30/60/90 plan, and each has a slide deck.

`ticket-analysis.md` — the symmetric reliability measurement across both platforms.

`workflow-gap.md` — the 5.5% / 94% volume split between Bravo's two generations.

Decision A has a firm answer and costs 18–33 engineer-days. Decision B is open, and the two measurements that would settle it cost days — and nobody has taken them.